

SEMINAR 3

Reprezentarea numerelor în calculator.

I] Reprezentarea numerelor întregi n biți

$$[x]_D = \begin{cases} 0 \overrightarrow{|x|}, & x \geq 0 \\ 1 \overrightarrow{|x|}, & x < 0 \end{cases}$$

Direct

$$[x]_D = \begin{cases} 0 \overrightarrow{|x|}, & x \geq 0 \\ 1 \overrightarrow{|x|}, & x < 0 \end{cases}$$

$$[x]_I = \begin{cases} 0 \overrightarrow{|x|}, & x \geq 0 \\ 1 \overrightarrow{\begin{matrix} |x| \\ 0 \uparrow \\ 1 \downarrow \end{matrix}}, & x < 0 \end{cases}$$

INVERS

$$[x]_I = \begin{cases} 0 \overrightarrow{|x|}, & x \geq 0 \\ 1 \overrightarrow{\begin{matrix} |x| \\ 0 \uparrow \\ 1 \downarrow \end{matrix}}, & x < 0 \end{cases}$$

$$[x]_C = \begin{cases} 0 & |x|, x \geq 0 \\ 1 & \begin{array}{c} |x| \\ \downarrow \downarrow \\ 0 \quad 1 \\ \downarrow \downarrow \\ 1 \quad 0 \end{array}, x < 0 \end{cases}$$

Complementar

$$[x]_C = \begin{cases} 0, |x|, x \geq 0 \\ 1, \begin{array}{c} |x| \\ \downarrow \downarrow \\ 0 \quad 1 \\ \downarrow \downarrow \\ 1 \quad 0 \end{array}, x < 0 \end{cases}$$

Exercitiu

$$n=8$$

$$X = 12^3_{(10)} = 1111 \oplus 11_{(2)}$$

$$[X]_A = \boxed{0 \mid 1 \mid 1 \mid 1 \mid 1 \mid 0 \mid 1 \mid 1}$$

$$[X]_Y = \boxed{0 \mid 1 \mid 1 \mid 1 \mid 1 \mid 0 \mid 1 \mid 1}$$

$$[X]_C = \boxed{0 \mid 1 \mid 1 \mid 1 \mid 1 \mid 0 \mid 1 \mid 1}$$

$$y = -327_{(10)} = -507_{(8)} = -101000111$$

$$\begin{array}{r|l} 327 & 8 \\ \hline 32 & 70 \\ \hline 0 & 70 \\ \hline 7 & 0 \end{array} \quad \begin{array}{r|l} 8 & \\ \hline 15 & \end{array}$$

$$[y]_D = [1|1|0|0|0|1|1|1]$$

$$[y]_J = [1|0|1|1|1|0|0|0]$$

$$[y]_C = [1|0|1|1|1|0|0|1]$$

$$z = 0,237_{(10)} = 0,171_{(8)} = 0,00111001_{(2)}$$

$m=8$ bits

$$\begin{array}{r|l} 0,237 & 8 \\ \hline 1,896 & \end{array} \quad \begin{array}{r|l} 0,171 & 8 \\ \hline 1,368 & \end{array} \quad \begin{array}{r|l} 0,168 & 8 \\ \hline 1,344 & \end{array}$$

$$[z]_D = [0|0|0|1|1|1|1|0]$$

$$[z]_J = [0|0|0|1|1|1|1|0]$$

$$[z]_C = [0|0|0|1|1|1|1|0]$$

Operații în cod complement

$$[x]_C \oplus [y]_C = \begin{cases} [x]_C + [y]_C & \text{dacă nu depășește n biți} \\ [x]_C + [y]_C & \text{dacă pe n biți, altfel transport răsare în CF} \\ & \text{în biți} \end{cases}$$

Dacă x și y au același semn $\rightarrow$ control cu semnul sumei lui $[x]_C \oplus [y]_C$,
avem depășire

Virgulă mobilă

$$x = +156,23_{(10)} = 10011100,001110_{(2)} \\ = 0,10011100001110_{(2)} \cdot 2^8$$

$$C = E + Q = 8 + 127 = 8 + 128 - 1 = 128 + 7 = \begin{array}{r} 10000000 + \\ \quad 111 \\ \hline 10000111_{(2)} \end{array}$$

0	10000111	10011100001110	101110000
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$$\begin{array}{r} 0,72 \\ \hline 8 \\ \hline 5,76 \end{array} \quad \begin{array}{r} 0,72 \quad 0,08 \\ \hline 8 \quad 8 \\ \hline 6,08 \quad 0,64 \end{array}$$

Mantisă supraunitară

$$y = -142,5_{(10)} = -10001110,1_{(2)} \\ = -1,00011101 \cdot 2^7$$

$$C = Q + E = 127 + 7 = 128 + 6 = 2^7 + 6 = \begin{array}{r} 10000000 + \\ \quad 110 \\ \hline 10000110 \end{array}$$

1	1000110	00011101	000000000000
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$$-\frac{3}{16} = -3 \cdot 16^{-1} = -013_{(16)} = -0, \overbrace{0011}_{(2)} = 0, \underline{11}_{(2)} \cdot 2^{-2}$$

$$C = Q + E = 127 - 2 = 1111111_{(2)} -$$

$$\begin{array}{r} 10 \\ \hline 1111101_{(2)} \end{array}$$

$$\left(1 \mid 01111101 \mid \color{red}{11} 0000000000 \dots \right)$$